

RelWeights, Graph Laplacians, and Embedding Transfer

A Structured Mathematical Reference Guide

SECTION 1 — INCIDENCE MATRICES

Binary incidence matrix:

$$B_{\text{bin}}(i,k) = 1 \text{ if } |\alpha_i \cap \beta_k| > 0 \text{ else } 0$$

Area-overlap incidence matrix:

$$B_{\text{area}}(i,k) = |\alpha_i \cap \beta_k| / |\alpha_i|$$

Row-standardization property:

$$\sum_k B_{\text{area}}(i,k) = 1$$

Binary incidence is support pattern of area-overlap incidence:

$$B_{\text{bin}}(i,k) = 1(B_{\text{area}}(i,k) > 0)$$

SECTION 2 — RELWEIGHTS MATRIX

Definition:

$$R = B B^T - \text{diag}(B B^T)$$

Entrywise:

$$R_{ij} = \sum_k B_{ik} B_{jk}$$

Interpretation:

shared-support similarity kernel between α -units

Properties:

$$R_{ij} \geq 0$$

R symmetric

$$\text{rank}(R) \leq \text{rank}(B)$$

R is positive semidefinite before diagonal removal

SECTION 3 — LAPLACIAN MATRIX

Definition:

$$L_R = D_R - R$$

where:

$$(D_R)_{ii} = \sum_j R_{ij}$$

Entrywise compact form:

$$(L_R)_{ij} = \delta_{ij} \sum_k R_{ik} - R_{ij}$$

Kronecker delta:

$$\delta_{ij} = 1 \text{ if } i=j \text{ else } 0$$

SECTION 4 — QUADRATIC FORM IDENTITY

Core identity:

$$x^T L_R x = (1/2) \sum_{\{i,j\}} R_{ij} (x_i - x_j)^2$$

Interpretation:

Relational roughness energy

SECTION 5 — POSITIVE SEMIDEFINITENESS

Since:

$$(x_i - x_j)^2 \geq 0$$

$$R_{ij} \geq 0$$

we have:

$$x^T L_R x \geq 0$$

Therefore:

all eigenvalues $\lambda \geq 0$

Hence:

$$\lambda_{\min} = 0$$

SECTION 6 — NULLSPACE PROPERTY

Constant vector:

$$\mathbf{1} = (1, 1, \dots, 1)^T$$

satisfies:

$$\mathbf{L}_R \mathbf{1} = 0$$

Thus:

constants have zero relational curvature

SECTION 7 — CONNECTED COMPONENT RESULT

Multiplicity of eigenvalue zero equals number of connected components in RelWeights graph

SECTION 8 — RAYLEIGH QUOTIENT

Definition:

$$R_L(x) = (x^T L_R x) / (x^T x)$$

Bounds:

$$0 \leq R_L(x) \leq \lambda_{\max}$$

SECTION 9 — GRAPH DIFFUSION INTERPRETATION

Discrete diffusion equation:

$$dx/dt = -L_R x$$

Interpretation:

information propagates across shared supports

SECTION 10 — RELATIONAL ENERGY FUNCTIONAL

Energy:

$$E(x) = x^T L_R x$$

Measures inconsistency with inherited-support geometry

SECTION 11 — EIGENVECTOR INTERPRETATION

Eigenproblem:

$$L_R u_k = \lambda_k u_k$$

Ordering:

$\lambda=0 \rightarrow$ constant mode

small $\lambda \rightarrow$ smooth variation

large $\lambda \rightarrow$ oscillatory variation

SECTION 12 — EMBEDDING TRANSFER MODEL

Baseline projection:

$$E_{\alpha}^{(0)} = B E_{\beta}$$

Regularized refinement:

$$E_{\hat{\alpha}} = \operatorname{argmin}_X \|X - E_{\alpha}^{(0)}\|_F^2 + \lambda \operatorname{tr}(X^T L_R X)$$

Closed-form solution:

$$E_{\hat{\alpha}} = (I + \lambda L_R)^{-1} E_{\alpha}^{(0)}$$

SECTION 13 — TRACE DECOMPOSITION

Identity:

$$\text{tr}(X^T L_R X) = \sum_j x_j^T L_R x_j$$

Interpretation:

columnwise relational energy

SECTION 14 — CONSERVATION PROPERTY

Since:

$$L_R \mathbf{1} = 0$$

Then:

$$(I + \lambda L_R)^{-1} \mathbf{1} = \mathbf{1}$$

Therefore:

embedding column sums preserved

SECTION 15 — KERNEL INTERPRETATION OF RELWEIGHTS

Kernel form:

$$R = B B^T$$

Thus:

$$R_{ij} =$$

RelWeights defines valid similarity kernel

SECTION 16 — LAPLACIAN AS DISCRETE CURVATURE OPERATOR

$$(L_R x)_i = \sum_j R_{ij} (x_i - x_j)$$

Discrete harmonic condition:

$L_R x = 0 \Rightarrow$ local relational averaging

SECTION 17 — LAPLACIAN ENERGY GEOMETRY

Energy expression:

$$x^T L_R x = (1/2) \sum_{\{i,j\}} R_{ij} (x_i - x_j)^2$$

Measures deviation from smooth support-consistent structure

SECTION 18 — NORMALIZED LAPLACIANS

Random-walk Laplacian:

$$L_{rw} = D^{-1} L$$

Symmetric normalized Laplacian:

$$L_{sym} = D^{-1/2} L D^{-1/2}$$

SECTION 19 — SPECTRAL BOUNDS

Eigenvalue inequality:

$$0 \leq x^T L_R x \leq \lambda_{\max} \|x\|^2$$

SECTION 20 — RELWEIGHTS VS ADJACENCY MATRICES

Adjacency:

similarity via physical borders

RelWeights:

similarity via shared inherited supports

SECTION 21 — AREA-OVERLAP VS BINARY INCIDENCE EFFECT

Binary:

counts shared supports

Area-based:

weights similarity by proportional overlap

SECTION 22 — MACHINE LEARNING INTERPRETATION

RelWeights kernel:

$$K_{\alpha} = B K_{\beta} B^T$$

Kernel transport across spatial supports

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